3. Estimation of oxalic acid

Aim :

To estimate the amount of oxalic acid dissolved in 500 ml of the given solution volumetrically. For this you are given with a standard solution of ferrous ammonium sulphate (FAS) of normality 0.1 N and potassium permanganate solution as link solution.

Principle:

During these titrations, oxalic acid is oxidized to CO_2 and MnO_4^- ions (from $KMnO_4$) is reduced to Mn^{2+} ion.

Oxidation : $\underset{\text{Pink}}{MnO_4^-} + 8H^+ + 5e^- \longrightarrow \underset{\text{colourless}}{Mn^{2+}} + 4H_2O$

Reduction : $\underset{\text{Pink}}{MnO_4^-} + 8H^+ + 5e^- \longrightarrow \underset{\text{colourless}}{Mn^{2+}} + 4H_2O$

Overall reaction : $5(COOH)_2 + 2MnO_4^- + 6H^+ \longrightarrow 10CO_2 + 2Mn^{2+} + 8H_2O$

Since one mole oxalic acid releases 2 moles of electrons, the equivalent weight of oxalic acid $= \frac{106}{2} = 63$ (oxalic acid is dihydrated)

Short procedure:

s.no	Content	Titration-I	Titration-II
1	Burette solution	$KMnO_4$	$KMnO_4$
2	Pipette solution	20 ml of standard FAS	20 ml of unknown oxalic acid
3	Acid to be added	20ml of 2N H_2SO_4(approx)	20ml of 2N H_2SO_4 (approx)
4	Temperature	Lab temperature	60 – 70 ^{0}C
5	Indicator	Self-indicator ($KMnO_4$)	Self-indicator ($KMnO_4$)
6	End point	Appearance of permanent pale pink colour	Appearance of permanent pale pink colour
7	Equivalent weight of oxalic acid = 63		

Procedure :

Titration–I

(Link $KMnO_4$)Vs (Standard FAS)

Burette is washed with water, rinsed with $KMnO_4$ solution and filled with same $KMnO_4$ solution up to the zero mark. Exactly 20 ml of standard FAS solution is pipetted out into the clean, washed conical flask. To this FAS solution, approximately 20ml of 2N sulphuric acid is added. This mixture is titrated against $KMnO_4$ Link solution from the burette. $KMnO_4$ is added drop wise till the appearance of permanent pale pink colour. Burette reading is noted and the same procedure is repeated to get concordant values.

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